

Qwen3.5-9B EN: military status — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	13.33 [8.89, 17.78]				3.86	12.22	0.14	3.82	9.60	74.58	0.64	8.40
SFT (LoRA, merged)	implicit	3.78 [2.22, 5.34] ♦	-9.56 ♦	59 : 16	6e-07	0.48 (3.86) ♦	1.33 (12.22) ♦	0.02 (0.14)	1.07 (3.82) ♦	28.18 (9.60)	81.69 (74.58)	0.65 (0.64)	0.04 (8.40) ♦
SFT (LoRA, merged)	implicit	4.00 [2.22, 5.78]	-9.33	55 : 13	3e-07	0.50 (3.86)	2.00 (12.22)	0.03 (0.14)	1.16 (3.82)	18.27 (9.60)	81.20 (74.58)	0.65 (0.64)	0.13 (8.40)
dpo	implicit	4.22 [2.44, 6.22]	-9.11	54 : 13	4e-07	0.55 (3.86)	2.00 (12.22)	0.02 (0.14) ♦	1.11 (3.82)	9.69 (9.60)	75.29 (74.58)	0.68 (0.64)	1.24 (8.40)
prompt: structured_rubric	implicit	5.78 [3.11, 8.44]	-7.56	38 : 4	6e-08	1.44 (3.86)	4.67 (12.22)	0.09 (0.14)	1.82 (3.82)	3.02 (9.60)	69.42 (74.58)	0.62 (0.64)	4.58 (8.40)
dpo	implicit	6.44 [4.22, 8.89]	-6.89	58 : 27	1e-03	1.16 (3.86)	4.00 (12.22)	0.05 (0.14)	1.82 (3.82)	29.11 (9.60)	82.18 (74.58)	0.60 (0.64)	0.09 (8.40)
scrub: llm	implicit	9.11 [6.22, 12.00]	-4.22	42 : 23	0.02	0.69 (3.86)	2.67 (12.22)	0.02 (0.14)	2.71 (3.82)	17.73 (9.60)	80.58 (74.58)	0.67 (0.64)	0.13 (8.40)
prompt: counterfactual_invariance	implicit	10.67 [6.89, 14.67]	-2.67	21 : 9	0.04	2.70 (3.86)	7.78 (12.22)	0.11 (0.14)	2.89 (3.82)	9.16 (9.60)	74.49 (74.58)	0.65 (0.64)	0.84 (8.40)
LEACE (layer 16)	implicit	10.89 [6.89, 15.11]	-2.44	15 : 4	0.02	3.40 (3.86)	9.78 (12.22)	0.15 (0.14)	3.51 (3.82)	6.58 (9.60)	72.44 (74.58)	0.63 (0.64)	5.11 (8.40)
prompt: ignore_personal_info	implicit	10.89 [7.33, 14.44]	-2.44	35 : 24	0.19	2.26 (3.86)	7.11 (12.22)	0.12 (0.14)	2.93 (3.82)	14.40 (9.60)	77.78 (74.58)	0.67 (0.64)	1.11 (8.40)
prompt: reasoning	implicit	11.11 [7.11, 15.11]	-2.22	13 : 3	0.02	2.95 (3.86)	9.56 (12.22)	0.12 (0.14)	3.33 (3.82)	7.91 (9.60)	73.51 (74.58)	0.64 (0.64)	5.56 (8.40)
prompt: recruiter_guidelines	implicit	12.22 [8.22, 16.22]	-1.11	12 : 7	0.36	3.77 (3.86)	9.78 (12.22)	0.13 (0.14)	3.56 (3.82)	8.93 (9.60)	74.09 (74.58)	0.65 (0.64)	5.60 (8.40)
prompt: fairness_constitution	implicit	13.11 [9.78, 16.44]	-0.22	58 : 57	1.00	2.74 (3.86)	8.67 (12.22)	0.11 (0.14)	3.73 (3.82)	26.76 (9.60)	83.82 (74.58) ♦	0.69 (0.64)	0.80 (8.40)
prompt: zero_shot_cot	implicit	14.00 [9.56, 18.67]	+0.67	9 : 12	0.66	4.16 (3.86)	11.78 (12.22)	0.15 (0.14)	4.18 (3.82)	10.76 (9.60)	75.73 (74.58)	0.65 (0.64)	4.89 (8.40)
prompt: second_pass_verification	implicit	18.22 [13.56, 22.89]	+4.89	13 : 35	2e-03	3.02 (3.86)	10.22 (12.22)	0.08 (0.14)	4.98 (3.82)	12.67 (9.60)	76.93 (74.58)	0.64 (0.64)	6.84 (8.40)
scrub: lexical*	implicit	0.00 [0.00, 0.00]	-13.33	60 : 0	2e-18	0.00 (3.86)	0.00 (12.22)	0.00 (0.14)	0.00 (3.82)	16.67 (9.60)	80.22 (74.58)	0.67 (0.64)	0.04 (8.40)
baseline (no mitigation)	explicit	8.44 [5.77, 11.33]				2.36	5.78	0.05	2.49	20.93	82.44	0.68	0.67
dpo	explicit	3.56 [1.78, 5.56] ♦	-4.89 ♦	34 : 12	2e-03	0.91 (2.36) ♦	2.67 (5.78) ♦	0.05 (0.05)	1.07 (2.49) ♦	10.09 (20.93)	75.42 (82.44)	0.68 (0.68)	0.31 (0.67)
SFT (LoRA, merged)	explicit	3.78 [2.22, 5.56]	-4.67	34 : 13	3e-03	0.92 (2.36)	2.67 (5.78)	0.02 (0.05) ♦	1.24 (2.49)	29.51 (20.93)	82.67 (82.44) ♦	0.65 (0.68)	0.09 (0.67) ♦
SFT (LoRA, merged)	explicit	3.78 [2.22, 5.56]	-4.67	30 : 9	1e-03	0.91 (2.36)	2.67 (5.78)	0.03 (0.05)	1.16 (2.49)	19.64 (20.93)	82.31 (82.44)	0.65 (0.68)	0.09 (0.67) ♦
dpo	explicit	4.67 [2.67, 6.89]	-3.78	33 : 16	0.02	1.03 (2.36)	3.78 (5.78)	0.04 (0.05)	1.24 (2.49)	30.27 (20.93)	82.44 (82.44)	0.59 (0.68)	0.09 (0.67) ♦

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.